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Experiences of health risks among medical waste handlers in low and middle-income countries: A systematic review protocol of qualitative studies

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Experiences of health risks among medical waste handlers in low and middle-income countries: A systematic review protocol of qualitative studies

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ABSTRACT

Introduction

Health risks associated with medical waste management (MWM) in low and middle-income countries (LMICs) are important for public health and research. Although studies of MWM and health risks among medical waste handlers in LMICs are available, literature remains under-synthesised and knowledge fragmented. This systematic review will provide a comprehensive synthesis of evidence regarding the factors that affect MWM and health risks experienced by medical waste handlers in LMICs. The review will answer the following questions: (i) what factors affect MWM in LMICs? and (ii) what are common health risks experienced by medical waste handlers in LMICs?

Methods and analysis

Seven specific electronic databases (e.g. Scopus, Ovid MEDLINE, EMBASE, Global Health, CINAHL, ProQuest, and PsycINFO) will be searched, and studies will be included based on inclusion criteria. Two authors will review the citations and full-texts, extract data, and complete the quality appraisal independently. A third reviewer will check discrepancies when a consensus cannot be reached on differences between the two reviewers. Data extraction will be conducted using the Joanna Briggs Institute standardised data extraction form for qualitative research. The quality of articles will be assessed using a Critical Appraisal Skills Programme (CASP) checklist. Results from eligible articles will be synthesised into a set of findings using the thematic framework analysis approach and will be reported according to the Enhancing Transparency in Reporting the Synthesis of Qualitative Research (ENTREQ) statement.

Ethics and dissemination

This protocol has been registered with the International Prospective Register of Systematic Reviews (PROSPERO), registration number CRD42020226851.

Keywords

Medical waste management; waste handlers; health risks; low and middle-income countries

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Strengths and limitation of this study

- The primary strength of the systematic review is that it will bring together the current knowledge in the area of MWM and health risks experienced by medical waste handlers in LMICs and develop a context-specific comprehensive understanding of factors that affect MWM and health risks experienced by medical waste handlers.
- To the best of our knowledge, this review will be the first evidence synthesis of factors that affect MWM and health risks experienced by medical waste handlers in LMICs using a systematic approach.
- This review will only include studies published in the English language. Excluding studies published in languages other than English may produce publication bias and impact generalizability.

BACKGROUND

The healthcare sector in low and middle-income countries (LMICs) is growing fast [1], due to rapid population growth and increased access to medical services [2], contributing to generating increased medical wastes[3]. According to the World Health Organisation (WHO) [4], medical wastes refer to wastes and by-products generated by healthcare and medical research settings, including medical or diagnostic centres and hospitals. While most medical wastes are not hazardous, around 15% of them is considered dangerous wastes [3 5]. These hazardous medical wastes may be infectious (contaminated blood, cultures and stocks of contagious agents), sharps (syringes, needles, disposable etc.), cytotoxic ((e.g. wastes comprising cytostatic drugs—commonly utilised in cancer therapy, genotoxic chemicals), radioactive (e.g. unused liquids from radiotherapy or laboratory research, contaminated glassware, packages, or absorbent paper; urine and excreta from patients treated), chemical (disinfectants, sterilants and heavy metals etc.) or pharmaceutical (expired, unused and contaminated drugs and vaccines) [4-7]. These wastes contain pathogenic micro-organisms that may enter into human bodies, especially in medical waste handlers in LMICs, through various routes, including punctures, abrasions or cuts in the skin, inhalation, and ingestion [3]. Thus, medical waste constituents have the potential for infection and injury.

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Health risk can be characterised as the probability of a situation or an event and its consequences related to infectious diseases (Hepatitis B and C, HIV, and COVID-19) and other health complications (e.g. respiratory disorders, cancer, burn, and skin irritation) of individuals' (e.g. waste handlers) health [8]. Medical waste handlers, particularly those in LMICs [9], are potentially at a higher health risk to various infections and injuries due to the manual sorting of dangerous materials at waste disposal sites [3]. Compared to high-income countries, medical waste management (MWM) in LMICs is not well-equipped with the resources and capability to reduce health risks of medical waste handlers. Additionally, the good MWM practices are not followed by waste handlers due to inadequate education and training [10]. Furthermore, the emergence of the COVID-19 pandemic has resulted in increased dangerous medical wastes from increased use of personal protective equipment (PPE) [11] such as gloves, surgical masks, goggles or face shields, gowns, and respirators (i.e. N95 or FFP2) and aprons [12]. Improper management of these used infectious medical wastes, especially in LMICs, may increase the risk of infections and injuries, including the COVID-19, Hepatitis B and C, and sharps-inflicted injuries, among medical waste handlers [13 14].

Although studies of MWM and its health risks among medical waste handlers in LMICs are available, the studies are scattered in answering different research questions, and the evidence on MWM and medical waste handlers' exposure to health risks in LMICs were not systematically synthesised. Identifying, collating, synthesising, and appraising available literature systematically will provide a comprehensive understanding of medical waste handlers' health risks and factors affecting MWM in various settings and help inform future health policy and practices related to MWM in LIMCs. Furthermore, systematically synthesising various findings of available studies may help LMICs, which may not follow the best available MWM practices and can learn from similar settings, thus improving their MWM. This sets a strong rationale for synthesising available evidence and understanding gaps through this review and suggests recommendations for improving MWM and reducing health risks. Therefore, this systematic review will identify, appraise and synthesise qualitative evidence on the factors that affect MWM in LMICs and explore common health risks experienced by medical waste handlers in LMICs. The review will answer the following questions: (i) what factors affect MWM in LMICs? and (ii) what are common health risks experienced by medical waste handlers in LMICs?

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METHODS

This review will utilise the ENTREQ statement to report the thematic synthesis of qualitative studies [15 16]. The ENTREQ statement aids investigators to report the phases most usually related to the synthesis of qualitative health research, including searching and selecting qualitative studies, quality appraisal, and methods for synthesising qualitative findings [15].

This review protocol was registered with the PROSPERO international prospective register of systematic reviews [CRD42020226851].

Search strategy

In this review, the following seven electronic databases will be searched for relevant studies in the past ten years: Scopus, Ovid MEDLINE, EMBASE, Global Health, CINAHL, ProQuest, and PsycINFO. In the case of relevant publications which might have been missed during the initial search, a further search of the bibliographical references of all eligible publications, complemented by citation tracking using Google Scholar will be conducted. A search strategy will be developed on MEDLINE and adapted to other databases as shown in Additional file 1. The search will apply subject heading truncations (*), and Boolean operators ("AND", "OR" and "NOT") depending on the specifications of the databases. A research librarian will be consulted to finalise the search strategy. A search log will be kept for accountability and transparency. Two authors (MNH and TGH) will identify relevant literature, run a screening using titles, and review abstracts during the literature search. Final selection will be accomplished by reviewing full texts and applying eligibility criteria to include studies aligned with the research objectives. Database searches will be re-run prior to the final analysis to ensure any recent publications are included in the review. The search will be limited to English literature only since we have limited or no financial and logistical capacity to retrieve and translate articles published in languages other than English.

The search strategy will use the following key terms for subject headings: ‘medical waste’, ‘waste management’, ‘medical waste handlers’, ‘health risks’, ‘infectious disease’, ‘health complications’, ‘low and middle-income countries’, ‘developing countries’ and ‘resource-limited countries’. Along with these key concepts for search criterion, specific diseases

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(including Covid-19/coronavirus, HIV/AIDs, Hepatitis B and C, cancer and health complications (e.g. respiratory disorders, cancer, burn, and skin irritation) will be searched to find relevant articles.

Inclusion and Exclusion Criteria

Types of studies

All qualitative studies published in peer-reviewed journals (1) between 2011 and 2021 with full texts available and accessible will be included in the review. Qualitative literature will be chosen because the review's interest is to understand better the experience of health risks of the medical waste handlers. Studies (2) that apply established qualitative data collection techniques (such as semi-structured and unstructured interviews, focus group discussions, direct observations, or semi-structured questionnaires that permit free texts; (3) highlight the health risks experienced by medical wastes handlers; (4) focus on waste management systems in LMICs; and (5) published in English will be included in the analysis. We will re-run the database searches prior to the final analysis.

Studies will be excluded if they (1) do not provide account/quotes of study participants, or (2) apply mixed-methods which lack accounts or quotes of study participants; (3) are study protocols, reviews, editorials, letters to editors, commentaries, conference abstract and posters??, and opinion pieces; and (4) focus on medical waste handlers having severe pre-existing health risks before they became involved in MWM.

Participants/populations

Participants will include medical waste handlers, defined as all workers (e.g. waste cleaners, waste pickers, collectors, recycling waste operators, scavengers, landfill workers, garbage workers) who are directly involved in medical waste collection and final disposal at waste disposal sites (city corporation bins and general landfill sites), medical research institutes, and healthcare facilities (public and private hospitals/clinics) as recommended by the WHO [7]. The participants will include: (1) medical waste handlers of all age groups, including children from 10 years and above; (2) both male and female garbage workers; (3) waste pickers who search medical waste before or after the final disposal; (4) scavengers who search medical wastes before or after the final disposal.

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Type of setting

We will include studies from LMICs where medical waste workers are engaged in collecting MW from waste disposal sites, medical research institutes, and healthcare facilities and disposing of waste at disposal sites.

Interventions

To ensure the applicability of our review, we will include studies that focus on the MWM interventions in the waste disposal and healthcare facilities in LMICs.

Outcomes

This review's primary outcome will consist of factors affecting MWM in LMICs. The secondary outcome will be the health risks experienced by the medical waste handlers in LMICs.

DATA COLLECTION AND ANALYSIS

Study selection

All studies identified in the search will be exported into the reference manager EndNote library. A three-step screening process will be applied to screen and select eligible studies. The first step will involve assessing the titles for relevance, in which case clearly irrelevant titles will be excluded. After that, abstract screening for eligibility and relevance will be conducted, and studies that do not meet the inclusion criteria will be excluded. In the final step, the full text of the retained studies will be further screened for inclusion in the review. Final studies selected from the full-text screening will be recorded. Two reviewers (MNH and TGH) will independently perform the search, and any disagreements will be resolved by a third reviewer (SH).

Data extraction and management

Data will be independently extracted by two review authors (MNH and TGH) from eligible studies onto the Joanna Briggs Institute standardised data extraction form for qualitative research [17] (see Additional file 2) and populated with variables pertaining to the study population and phenomena of interest. If there are disagreements between the two

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authors, a third author (SH) will double-check and verify the differences. Study characteristics that will be extracted will include the first author's name and publication year, data collection period and country in which the study was conducted. Then, descriptive data will be captured, including the study design, study population, sample size, sampling procedures and data collection procedures. The main findings and accounts of participants that explain the factors affecting MWM and the common health risks experienced by medical waste handlers will be systematically extracted.

Quality appraisal

Studies included in this review will be assessed for methodological quality using the Critical Appraisal Skills Programme (CASP) quality assessment tool for qualitative studies [18]. The CASP tool consists of 10 questions: 9 addressing 'quality' and 1 addressing 'value' (contribution to existing literature). Two reviewers (MNH and TGH) will score the quality appraisal assessment independently. We will apply both scoring methods and discussion to arrive at a consensus on quality. A third author (BA) will adjudicate should no agreement be reached between the two reviewers.

In line with other qualitative reviews [19 20], this review will not exclude studies on the basis of quality. Instead, a further understanding of the contributions of the included studies will be provided at a later stage of this review. Regardless of the quality appraisal score, all studies relevant to the review questions will undergo data extraction and synthesis to assess and compare findings. The results of the quality appraisal will be reported in a narrative form and a table.

DATA SYNTHESIS AND ANALYSIS

In accordance with existing systematic reviews [21 22], we will adopt the thematic framework analysis approach to identify the factors that affect MWM and common health risks from the narratives of critical findings in selected studies [23]. The thematic framework analysis is one of the approaches recommended by the Cochrane Qualitative Review Methods Group [24] to perform syntheses of qualitative studies [25]. Thematic synthesis is appropriate where the evidence is likely to be mainly descriptive [21] and will enrich our understanding of health risks experienced by medical waste handlers and factors that can facilitate or prevent

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MWM. We will follow five stages of the thematic framework analysis to synthesise the data [21 26] (see below).

Familiarisation with the data: The first author (MNH) will begin with familiarising the data against the review aims and documenting recurrent themes across the studies.

Identifying a thematic framework: Given our interest in identifying priori themes (factors affecting MWM and common health risks experience), we will use a pre-determined thematic framework which was developed using literature (see Additional file 3) to guide the thematic analysis instead of developing our own priori framework. We will also explore emergent issues (including author, country, research focus etc.) related to review objectives. This thematic framework provides a detailed list of factors affecting MWM and health risk experiences of medical waste handlers.

Coding: The two authors (MNH and TGH) will independently read the extracted information to search for themes according to a pre-determined thematic framework and additional emergent themes. The framework will be revised as new themes emerge. All studies will be read until there are no new emerging themes. Coding of the data will be performed based on the themes identified in the data. Each primary study will be indexed using the codes related to the themes of the framework. Where appropriate, parts of the studies may be indexed with one or more codes. All agreements and disparities between the coders will be resolved through discussion and consensus among all authors.

Charting: We will group the data into descriptive themes that capture and describe patterns in the data across studies and present the themes in the form of an analysis table (chart). The columns and rows of the table will reflect the studies and related themes and enable us to compare findings of the studies across different themes and subthemes.

Mapping and interpretation: We will use charts to define the identified concepts and map the range and nature of the phenomena. This systematic review will explore relationships between the themes to help clarify the findings. We will synthesise findings across studies

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and interpret their meanings concerning the review research questions and emerging themes.

Investigation of heterogeneity

We will assess heterogeneity of findings by conducting subgroup analyses according to: medical waste handlers at various settings (health care facilities, incineration and waste disposal facilities, public and private healthcare facilities), ages (children versus adults), types of hazardous medical wastes (infectious, sharp, cytotoxic, radioactive, chemical, and pharmaceutical) and health risks (infectious versus non-infectious). We will also consider other axes that can emerge as essential when synthesising the evidence, such as settings (low-income versus middle-income countries) and waste management in urban and rural areas.

Patient and public involvement

No patient involved.

DISCUSSION

Globally, most LMICs have inadequate and inappropriate MWM practices, which exacerbate MWM-induced health risks among medical waste handlers [10 27]. Medical waste handlers' vulnerability to health risks and the necessity for improving MWM in LMICs highlight a research and policy priority. A substantial pool of literature exists on the subject matter but remains under-synthesised. The considerable body of literature supports the necessity for a review to provide a robust summary of evidence that could be drawn upon to optimise policies and planning related to MWM and health risks experienced by medical waste handlers in LMICs. By analysing the account of participants across their exposure settings, and demography, this review could provide a direction where possible future interventions or research should focus on reducing the risks and long-term effects on medical waste handlers. The results of this systematic review could be used to understand common health risks medical waste handlers often experienced in the LMICs and factors that played a role in facilitating or preventing the repercussions of such exposure. The review findings will inform and guide healthcare authorities on developing MWM-related interventions to improve MWM and reduce health risks among medical waste handlers in LMICs.

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ETHICS AND DISSEMINATION

Findings from this review will be published in a reputable peer-reviewed journal and presented at a relevant public health conference. This review does not require ethical approval, given that there is no collection of primary data. The protocol has been registered with the International Prospective Register of Systematic Reviews (PROSPERO), registration number CRD42020226851.

Contributors

MNH, SK, and AF conceptualised the idea and designed the review. MNH, JSM, SK, SH, and AF drafted the initial protocol. MNH and SK developed the search strategy. MNH interpreted, analysed and revised the manuscripts with intellectual input from TGH, SZH, BAI, RMI and AR. All authors read and approved the manuscript.

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Competing interests or Conflict of interests

None declared

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Search strategy

Table 1. Search terms and databases used in this systematic review.

#	Databases	Scopus, Ovid MEDLINE, EMBASE, Global Health, CINAHL, ProQuest, and PsycINFO.
1	Concept 1: Medical waste	(“Medical waste” OR "infectious waste" OR "hazardous waste" OR "communicable waste" OR "contagious waste" OR "dangerous waste" OR “risky waste”, OR “unsafe waste” OR "transmittable waste” OR “epidemic waste” OR “spreading waste” OR “biomedical waste” OR “clinical waste” OR “bio hazardous waste” OR “healthcare waste” OR “sharp waste*” OR “radioactive waste” OR “pathological waste *” OR “pharmaceutical waste” OR “laboratory waste” OR “chemical waste”) AND
2	Concept 2: Waste management	(“Waste manage* OR “waste disposal* OR “waste regulat*” OR “waste separate*”, OR “waste stor*”, OR “waste transport*” "waste intervention*” OR “waste process*" OR "waste handling*" OR " waste discharg*" OR “waste recycle*”) AND
3	Concept 3: Waste handlers	(“waste handler” "waste picker*" OR "garbage worker*" OR "rag picker*" OR scavenger*" OR " waste cleaner*" OR "landfill worker*" OR "informal worker*" OR "recycler*" OR "sanitation worker*" OR "trash collector*" OR “waste operat*” OR "sewerage worker*") AND

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4	Concept 4: Health risks	(“Health risk*” OR “health effect*” OR “health complication*” OR “communicable disease*” OR “infectious disease*” OR “contagious disease*” OR “contagious illness*” OR “transmissible disease” OR “physical risk” OR “respiratory disorder*” OR “liver disease” OR COVID-19 OR coronavirus OR cancer OR HIV OR AIDS OR hepatitis OR typhoid OR burn OR irritation, OR injury) AND
5	Concept 5: Low and middle-income countries	(“Low-income countr*” OR “middle-income countr*” OR “low and middle-income countr*” OR “resource-constrained countr*" “resource-limited countr*” OR “developing countr*" OR “marginalised countr*" OR “underprivileged countr*” “poor countr” OR “unindustrialised countr*” OR “emerging countr**” OR “less developed countr*” OR “underdeveloped countr*”) AND
6	Boolean operators and filters	#1 AND #2 AND #3 AND #4 AND #5 Filters: Full text; English, NOT systematic review, NOT quantitative studies.

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JBI QARI Data Extraction Tool for Qualitative Research

Reviewer _____ Date _____

Author _____ Year _____

Journal _____ Record Number _____

Study Description

Methodology

Method

Phenomena of interest

Setting

Geographical

Cultural

Participants

Data analysis

Authors conclusions

Comments

Complete Yes ☐ No ☐

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Findings	Illustration form Publication (page number)	Evidence		
		Unequivocal	Credible	Unsupported

Extraction of findings complete Yes ☐ No ☐

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only

Thematic framework analysis format for summarizing factors affecting MWM and health risks experienced by medical waste handlers in LMICs.

Main themes	Emerging sub-themes	Studies
Factors affecting MWM in LMICs		
Health risks experience		

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Primary Subject Heading:	Occupational and environmental medicine
Secondary Subject Heading:	Infectious diseases, Public health, HIV/AIDS, Health policy, Global health
Keywords:	Health & safety < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Public health < INFECTIOUS DISEASES, OCCUPATIONAL & INDUSTRIAL MEDICINE, COVID-19, Risk management < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Infection control < INFECTIOUS DISEASES

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ABSTRACT

Introduction

Health risks associated with medical waste management (MWM) in low and middle-income countries (LMICs) are an important public health concern. Although studies of MWM and health risks among medical waste handlers in LMICs are available, literature remains under-synthesised and knowledge fragmented. This systematic review will provide a comprehensive synthesis of evidence regarding the factors that influence MWM and health risks experienced by medical waste handlers in LMICs.

Methods and analysis

All qualitative studies published in peer-reviewed journals between 1 July 2011 and 30 June 2021 with full texts available and accessible will be included in the review. Seven specific electronic databases (e.g. Scopus, Ovid MEDLINE, EMBASE, Global Health, CINAHL, ProQuest, and PsycINFO) will be searched. Two authors will review the citations and full-texts, extract data, and complete the quality appraisal independently. A third reviewer will check discrepancies when a consensus cannot be reached on differences between the two reviewers. Data extraction will be conducted using the Joanna Briggs Institute standardised data extraction form for qualitative research. The quality of articles will be assessed using a Critical Appraisal Skills Programme (CASP) checklist. Results from eligible articles will be synthesised into a set of findings using the thematic framework analysis approach and will be reported according to the Enhancing Transparency in Reporting the Synthesis of Qualitative Research (ENTREQ) statement.

Ethics and dissemination

Findings from this review will be published in a peer-reviewed journal and presented at relevant public health conferences. This review is based on published articles, which does not require ethical approval, because there is no collection of primary data. This protocol has been registered with the International Prospective Register of Systematic Reviews (PROSPERO), registration number CRD42020226851.

Keywords

Medical waste management; waste handlers; health risks; low and middle-income countries

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Strengths and limitation of this study

- The primary strength of the systematic review is that it will bring together the current knowledge in the area of MWM and health risks experienced by medical waste handlers in LMICs and develop a context-specific comprehensive understanding of factors that influence MWM and health risks experienced by medical waste handlers.
- To the best of our knowledge, this review will be the first evidence synthesis of factors that influence MWM and health risks experienced by medical waste handlers in LMICs using a systematic approach.
- This review will only include studies published in the English language. Excluding studies published in languages other than English may produce publication bias and impact generalizability.

BACKGROUND

The healthcare sector in low and middle-income countries (LMICs) is growing fast [1] due to rapid population growth, and concomitant increased use of medical services [2] generates increased medical wastes[3]. Low-income countries are those with a gross national income (GNI) per capita, calculated using the World Bank Atlas method, of $1,045 or less in 2020. Lower-middle and upper-middle-income countries are those with a GNI per capita from $1,046 to $4,095 and $4,096 to $12,695, respectively [4]. According to the World Health Organisation (WHO) [5], medical wastes refer to wastes and by-products generated by healthcare and medical research settings, including medical or diagnostic centres and hospitals. While most medical wastes are not hazardous, around 15% of them is considered dangerous wastes [3 6]. These hazardous medical wastes may be infectious (contaminated blood, cultures and stocks of contagious agents), sharps (syringes, needles, disposable etc.), cytotoxic (e.g. wastes comprising cytostatic drugs—commonly utilised in cancer therapy, genotoxic chemicals), radioactive (e.g. unused liquids from radiotherapy or laboratory research, contaminated glassware, packages, or absorbent paper; urine and excreta from patients treated), chemical (disinfectants, sterilants and heavy metals etc.) or pharmaceutical (expired, unused and contaminated drugs and vaccines) [5-8]. These wastes

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contain pathogenic micro-organisms that may enter into human bodies, especially in medical waste handlers in LMICs, through various routes, including punctures, abrasions or cuts in the skin, inhalation, and ingestion [3]. Thus, medical waste constituents have the potential for injury and infection.

Health risk can be characterised as the probability of a situation or an event and its consequences related to infectious diseases (Hepatitis B and C, HIV, and COVID-19) and other health complications (e.g. respiratory disorders, cancer, burn, and skin irritation) of individuals' (e.g. waste handlers) health [9]. Medical waste handlers, particularly those in LMICs [10], are potentially at a higher health risk to various injuries and infections due to the manual sorting of dangerous materials at waste disposal sites [3]. Compared to high-income countries, medical waste management (MWM) in LMICs is not well-equipped with the resources and capability to reduce health risks of medical waste handlers. Additionally, the good MWM practices are not followed by waste handlers due to inadequate education and training [11]. Furthermore, the emergence of the COVID-19 pandemic has resulted in increased dangerous medical wastes from increased use of personal protective equipment (PPE) [12] such as gloves, surgical masks, goggles or face shields, gowns, and respirators (i.e. KN95 or FFP2) and aprons [13]. Improper management of these used infectious medical wastes, especially in LMICs, may increase the risk of injuries and infections, including the COVID-19, Hepatitis B and C, and sharps-inflicted injuries, among medical waste handlers [14] [15].

Although studies of MWM and its health risks among medical waste handlers in LMICs are available [16-18], the findings of these studies were not systematically synthesised. Identifying, collating, synthesising, and appraising available literature systematically will provide a comprehensive understanding of medical waste handlers' health risks and factors influencing MWM in various settings and help inform future health policy and practices related to MWM in LIMCs. Furthermore, systematically synthesising various findings of available studies may help LMICs, which may not follow the best available MWM practices and can learn from similar settings, thus improving their MWM. This sets a strong rationale for synthesising available evidence and understanding gaps through this review and

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suggests recommendations for improving MWM and reducing health risks. Therefore, this systematic review will identify, appraise and synthesise qualitative evidence on the factors that influence MWM in LMICs and explore common health risks experienced by medical waste handlers in LMICs. The review will answer the following questions: (i) what are the factors influencing MWM in LMICs? and (ii) what are common health risks experienced by medical waste handlers in LMICs?

METHODS

This protocol is informed by the standard Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) reporting guidelines [19] (see Additional file 1). This review will utilise the ENTREQ statement to report the thematic synthesis of qualitative studies [20 21]. The ENTREQ statement aids investigators to report the phases most usually related to the synthesis of qualitative health research, including searching and selecting qualitative studies, quality appraisal, and methods for synthesising qualitative findings [20].

Patient and public involvement

No patient involved.

Search strategy

In this review, the following seven electronic databases will be searched for relevant studies in the past ten years: Scopus, Ovid MEDLINE, EMBASE, Global Health, CINAHL, ProQuest, and PsycINFO. In the case of relevant publications which might have been missed during the initial search, a further search of the bibliographical references of all eligible publications, complemented by citation tracking using Google Scholar will be conducted. A search strategy will be developed on MEDLINE and adapted to other databases as shown in Additional file 2. The search will apply subject heading truncations (*), and Boolean operators ("AND", "OR" and "NOT") depending on the specifications of the databases. A research librarian will be consulted to finalise the search strategy. A search log will be kept for accountability and transparency. Two authors (MNH and TGH) will identify relevant literature, run a screening using titles, and review abstracts during the literature search. Final selection will be accomplished by reviewing full texts and applying eligibility criteria to

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include studies aligned with the research objectives. Database searches will be re-run prior to the final analysis to ensure any recent publications are included in the review. The search will be limited to English literature only since we have limited or no financial and logistical capacity to retrieve and translate articles published in languages other than English.

The search strategy will use the following key terms for subject headings: 'medical waste', 'waste management', 'medical waste handlers', 'health risks', 'infectious disease', 'health complications', 'low and middle-income countries', 'developing countries' and 'resource-limited countries'. Along with these key concepts for search criterion, specific diseases (including Covid-19/coronavirus, HIV/AIDs, Hepatitis B and C, cancer and health complications (e.g. respiratory disorders, cancer, burn, and skin irritation) will be searched to find relevant articles.

Inclusion and Exclusion Criteria

Types of studies

All qualitative studies published in peer-reviewed journals (1) between 1 July 2011 and 30 June 2021 with full texts available and accessible will be included in the review. Qualitative literature will be chosen because the review's interest is to understand better the experience of health risks of the medical waste handlers. Studies (2) that apply established qualitative data collection techniques (such as semi-structured and unstructured interviews, focus group discussions, direct observations, or semi-structured questionnaires that permit free texts; (3) highlight the health risks experienced by medical wastes handlers; (4) focus on waste management systems in LMICs; and (5) published in English will be included in the analysis.

Studies will be excluded if they (1) do not provide account/quotes of study participants, or (2) apply mixed-methods which lack accounts or quotes of study participants; and (3) are study protocols, reviews, editorials, letters to editors, commentaries, conference abstract, posters, and opinion pieces.

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Participants/populations

Participants will include medical waste handlers, defined as all workers (e.g. waste cleaners, waste pickers, collectors, recycling waste operators, scavengers, landfill workers, garbage workers) who are directly involved in medical waste collection and final disposal at waste disposal sites (city corporation bins and general landfill sites), medical research institutes, and healthcare facilities (public and private hospitals/clinics) as recommended by the WHO [8]. The participants will include: (1) medical waste handlers of all age groups, including children from 10 years and above; (2) both male and female garbage workers; (3) waste pickers who search medical waste before or after the final disposal; (4) scavengers who search medical wastes before or after the final disposal. However, this review will exclude medical waste handlers having severe pre-existing health risks before they became involved in MWM.

Type of setting

We will include studies from LMICs where medical waste workers are engaged in collecting MW from waste disposal sites, medical research institutes, and healthcare facilities and disposing of waste at disposal sites.

The following table summarises the inclusion and exclusion criteria for this systematic review.

Table 1. The inclusion and exclusion criteria for this study

Items	Inclusion criteria	Exclusion criteria
Topics	MWM systems in LMICs.	MWM systems, factors influencing MWM and health risks experienced by medical wastes handlers in upper-middle and high-income countries.
	Factors influencing MWM in LMICs.	
	Health risks experienced by medical wastes handlers.	
Types of studies	Studies that apply semi-structured and unstructured interviews, focus group discussions, direct observations, or semi-structured questionnaires which permit free texts.	Protocols, reviews, editorials, letters to editors, commentaries, conference abstract and posters, and opinion pieces.
		Studies that apply mixed-methods which lack accounts or quotes of study participants or do not provide account/quotes of study participants.
Participants	Medical waste handlers of all age groups, including children from 10 years and above; (2) both male and female	Children below 10 years of age.
		Medical waste handlers having severe pre-existing

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	garbage workers; (3) waste pickers who search medical waste before or after the final disposal; (4) scavengers who search medical wastes before or after the final disposal.	health risks before they became involved in MWM.
Settings	Waste disposal sites, medical research institutes, and healthcare facilities and disposal sites in LMICs.	Upper-middle and high-income countries.
Date range	1 July 2011 - 30 June 2021	Published prior to July 2011
Language	English	Excluded studies published in languages other than English

Outcomes

This review's primary outcome will consist of factors influencing MWM in LMICs. The secondary outcome will be the health risks experienced by the medical waste handlers in LMICs.

DATA COLLECTION AND ANALYSIS

Study selection

All studies identified in the search will be exported into the reference manager EndNote library. A three-step screening process will be applied to screen and select eligible studies. The first step will involve assessing the titles for relevance, in which case clearly irrelevant titles will be excluded. After that, abstract screening for eligibility and relevance will be conducted, and studies that do not meet the inclusion criteria will be excluded. In the final step, the full text of the retained studies will be further screened for inclusion in the review. Final studies selected from the full-text screening will be recorded. Two reviewers (MNH and TGH) will independently perform the study selection, and any disagreements will be resolved by a third reviewer (SH).

Data extraction and management

Data will be independently extracted by two review authors (MNH and TGH) from eligible studies onto the Joanna Briggs Institute standardised data extraction form for qualitative research [22] (see Additional file 3) and populated with variables pertaining to the study population and phenomena of interest. If there are disagreements between the two

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authors, a third author (SH) will double-check and verify the differences. Study characteristics that will be extracted will include the first author's name and publication year, data collection period and country in which the study was conducted. Then, descriptive data will be captured, including the study design, study population, sample size, sampling procedures and data collection procedures. The main findings and accounts of participants that explain the factors influencing MWM and the common health risks experienced by medical waste handlers will be systematically extracted.

Quality appraisal

Studies included in this review will be assessed for methodological quality and risk of bias [23] using the Critical Appraisal Skills Programme (CASP) quality assessment tool for qualitative studies [24]. The CASP tool consists of 10 questions: 9 addressing 'quality' and 1 addressing 'value' (contribution to existing literature). All included studies will be appraised as high, medium, or low quality, and the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach [25] will be utilised to assess the overall quality of the studies included in this review. Two reviewers (MNH and TGH) will score the quality appraisal assessment independently. We will apply both scoring methods and discussion to arrive at a consensus on quality. A third author (BA) will adjudicate should no agreement be reached between the two reviewers.

In line with other qualitative reviews [26 27], this review will not exclude studies on the basis of quality. Instead, a further understanding of the contributions of the included studies will be provided at a later stage of this review. Regardless of the quality appraisal score, all studies relevant to the review questions will undergo data extraction and synthesis to assess and compare findings. The results of the quality appraisal will be reported in a narrative form and a table.

DATA SYNTHESIS AND ANALYSIS

In accordance with existing systematic reviews [28 29], we will adopt the thematic framework analysis approach to identify the factors that influencing MWM in LMICs and common health risks from the narratives of critical findings in selected studies [30]. The thematic framework analysis is one of the approaches recommended by the Cochrane Qualitative

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Review Methods Group [31] to perform syntheses of qualitative studies [32]. Thematic synthesis is appropriate where the evidence is likely to be mainly descriptive [28] and will enrich our understanding of health risks experienced by medical waste handlers and factors that can facilitate or prevent MWM. We will follow five stages of the thematic framework analysis to synthesise the data [28 33] (see below).

Familiarisation with the data: The first author (MNH) will begin with familiarising the data against the review aims and documenting recurrent themes across the studies.

Identifying a thematic framework: Given our interest in identifying priori themes (factors influencing MWM and common health risks experience), we will use a pre-determined thematic framework which was developed using literature (see Additional file 4) to guide the thematic analysis instead of developing our own priori framework. We will also explore emergent issues (including author, country, research focus etc.) related to review objectives. This thematic framework provides a detailed list of factors influencing MWM and health risk experiences of medical waste handlers in LMICs.

Coding: The two authors (MNH and TGH) will independently read the extracted information to search for themes according to a pre-determined thematic framework and additional emergent themes. The framework will be revised as new themes emerge. All studies will be read until there are no new emerging themes. Coding of the data will be performed based on the themes identified in the data. Each primary study will be indexed using the codes related to the themes of the framework. Where appropriate, parts of the studies may be indexed with one or more codes. All agreements and disparities between the coders will be resolved through discussion and consensus among all authors.

Charting: We will group the data into descriptive themes that capture and describe patterns in the data across studies and present the themes in the form of an analysis table (chart). The columns and rows of the table will reflect the studies and related themes and enable us to compare findings of the studies across different themes and subthemes.

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Mapping and interpretation: We will use charts to define the identified concepts and map the range and nature of the phenomena. This systematic review will explore relationships between the themes to help clarify the findings. We will synthesise findings across studies and interpret their meanings concerning the review research questions and emerging themes.

Investigation of heterogeneity

We will assess heterogeneity of findings by conducting subgroup analyses according to: medical waste handlers at various settings (health care facilities, incineration and waste disposal facilities, public and private healthcare facilities), ages (children versus adults), types of hazardous medical wastes (infectious, sharp, cytotoxic, radioactive, chemical, and pharmaceutical) and health risks (infectious versus non-infectious). We will also consider other axes that can emerge as essential when synthesising the evidence, such as settings (low-income versus middle-income countries) and waste management in urban and rural areas.

DISCUSSION

Globally, most LMICs have inadequate and inappropriate MWM practices, which exacerbate MWM-induced health risks among medical waste handlers [11 34]. Medical waste handlers' vulnerability to health risks and the necessity for improving MWM in LMICs highlight a research and policy priority. A substantial pool of literature exists on the subject matter but remains under-synthesised. The considerable body of literature supports the necessity for a review to provide a robust summary of evidence that could be drawn upon to optimise policies and planning related to MWM and health risks experienced by medical waste handlers in LMICs. By analysing the account of participants across their exposure settings, and demography, this review could provide a direction where possible future interventions or research should focus on reducing the risks and long-term effects on medical waste handlers. The results of this systematic review could be used to understand common health risks medical waste handlers often experienced in the LMICs and factors that played a role in facilitating or preventing the repercussions of such exposure. The review findings will inform and guide healthcare authorities on developing MWM-related interventions to improve MWM and reduce health risks among medical waste handlers in LMICs.

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ETHICS AND DISSEMINATION

Findings from this review will be published in a reputable peer-reviewed journal and presented at a relevant public health conference. This review is based on published studies, which does not require ethical approval, because there is no collection of primary data. The protocol has been registered with the International Prospective Register of Systematic Reviews (PROSPERO), registration number CRD42020226851.

Contributors

MNH, SK, and AF conceptualised the idea and designed the review. MNH, JSM, SK, SH, and AF drafted the initial protocol. MNH and SK developed the search strategy. MNH interpreted, analysed and revised the manuscripts with intellectual input from TGH, SZH, BAI, RMI and AR. All authors read and approved the manuscript.

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Competing interests or Conflict of interests

None declared

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Additional file 1: PRISMA-P Statement—Checklist of items for qualitative studies

Items	Item numbers	Page numbers of manuscript
Title		
Identification	1a	1
Update	1b	N/A
Registration	2	2, 12
Authors		
Contact	3a	1
Contributions	3b	12
Amendments	4	N/A
Support		
Sources	5a	12
Sponsor	5b	12
Role of sponsor or funder	5c	12
Introduction		
Rationale	6	3-5
Objectives	7	2, 5-8
Methods		
Eligibility criteria	8	6-8
Information sources	9	1, 2, 5, 6, 12,
Search strategy	10	5-6
Study records		
Data management	11a	8-9
Selection process	11b	2, 8-10
Data collection process	11c	8-10
Data items	12	5-8, 12
Outcomes and prioritization	13	8
Risk of bias in individual studies	14	9
Data synthesis	15a-15d	9-11
Meta-biases	16	3, 6, 8
Confidence in cumulative evidence	17	9

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Search strategy

Additional file 2: Search terms used in this systematic review.

#	Search terms
1	(“Medical waste” OR "infectious waste" OR "hazardous waste" OR "communicable waste" OR "contagious waste" OR "dangerous waste" OR “risky waste”, OR “unsafe waste” OR "transmittable waste” OR “epidemic waste” OR “spreading waste” OR “biomedical waste” OR “clinical waste” OR “bio hazardous waste” OR “healthcare waste” OR “sharp waste*” OR “radioactive waste” OR “pathological waste *” OR “pharmaceutical waste” OR “laboratory waste” OR “chemical waste”)
2	(“Manage* OR “disposal* OR “regulat*” OR “separate*”, OR “stor*”, OR “transport*” " intervention*” OR “process*" OR " handling*" OR " discharg*" OR “recycle*”)
3	(Factor* OR determinant* OR issue* OR reason* OR cause*)
4	(“waste handler” "waste picker*" OR "garbage worker*" OR "rag picker*" OR scavenger*" OR " waste cleaner*" OR "landfill worker*" OR "informal worker*" OR "recycler*" OR "sanitation worker*" OR "trash collector*" OR “waste operat*” OR "sewerage worker*")
5	(“Health risk*” OR “health effect*” OR “health complication*” OR “communicable disease*” OR “infectious disease*” OR “contagious disease*” OR “contagious illness*” OR “transmissible disease” OR “physical risk” OR “respiratory disorder*” OR “liver disease” OR COVID-19 OR coronavirus OR cancer OR HIV OR AIDS OR hepatitis OR typhoid OR burn OR irritation, OR injury)
6	(“Low-income countr*” OR “middle-income countr*” OR “low and middle-income countr*” OR “resource-constrained countr*" “resource-limited countr*” OR “developing countr*" OR “marginalised countr*" OR “underprivileged countr*” “poor countr” OR “unindustrialised countr*” OR “emerging countr**” OR “less developed countr*” OR “underdeveloped countr*”)
7	#1 AND #2 AND #3 AND #6 Filters: Full text; English, NOT systematic review, NOT quantitative studies.
8	#1 AND #4 AND #5 AND #6 Filters: Full text; English, NOT systematic review, NOT quantitative studies.

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Additional file 3: JBI data extraction tool

JBI QARI Data Extraction Tool for Qualitative Research

Reviewer _____ Date _____

Author _____ Year _____

Journal _____ Record Number _____

Study Description

Methodology

Method

Phenomena of interest

Setting

Geographical

Cultural

Participants

Data analysis

Authors conclusions

Comments

Complete Yes ☐ No ☐

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Findings	Illustration form Publication (page number)	Evidence		
		Unequivocal	Credible	Unsupported

Extraction of findings complete Yes ☐ No ☐

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Additional file 4: Thematic framework analysis format for summarizing factors influencing MWM and health risks experienced by medical waste handlers in LMICs.

Main themes	Emerging sub-themes	Studies
Factors influencing MWM in LMICs		
Health risks experience		

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Medical waste management-related factors affecting health and experiences of health risks among medical waste handlers in low-and middle-income countries: A systematic review protocol of qualitative studies

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Medical waste management-related factors affecting health and experiences of health risks among medical waste handlers in low-and middle-income countries: A systematic review protocol of qualitative studies

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ABSTRACT

Introduction

Medical waste management (MWM)-related factors affecting the health of medical waste handlers (MWHs) and their health risks in low-and middle-income countries (LMICs) are an important public health concern. Although studies of MWM-related factors and health risks among MWHs in LMICs are available, literature remains under-synthesised and knowledge fragmented. This systematic review will provide a comprehensive synthesis of evidence regarding the individual, system, and policy-levels MWM-related factors that affect MWHs' health and their experiences of health risks in LMICs.

Methods and analysis

All qualitative studies published in peer-reviewed journals between 1 July 2011 and 30 June 2021 with full texts available and accessible will be included in the review. Seven specific electronic databases (e.g. Scopus, Ovid MEDLINE, EMBASE, Global Health, CINAHL, ProQuest, and PsycINFO) will be searched. Two authors will review the citations and full-texts, extract data, and complete the quality appraisal independently. A third reviewer will check discrepancies when a consensus cannot be reached on differences between the two reviewers. Data extraction will be conducted using the Joanna Briggs Institute standardised data extraction form for qualitative research. The quality of articles will be assessed using a Critical Appraisal Skills Programme (CASP) checklist. Results from eligible articles will be synthesised into a set of findings using the thematic framework analysis approach and will be reported according to the Enhancing Transparency in Reporting the Synthesis of Qualitative Research (ENTREQ) statement.

Ethics and dissemination

This review is based on published articles, which does not require ethical approval because there is no collection of primary data. Findings from this review will be published in a peer-reviewed journal and presented at relevant public health conferences. This protocol has been registered with the International Prospective Register of Systematic Reviews (PROSPERO), registration number CRD42020226851.

Keywords

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Medical waste management; waste handlers; health risks; low-and middle-income countries

Strengths and limitation of this study

- The primary strength of the systematic review is that it will bring together the current knowledge in the area of MWM-related factors affecting the health of medical waste handlers (MWHs) and their experiences of health risks in LMICs. This review will develop a context-specific comprehensive understanding of the individual, system, and policy-levels MWM-related factors that affect the health of MWHs, and their experiences of health risks.
- To the best of our knowledge, this review will be the first evidence synthesis of the individual, system, and policy-levels MWM-related factors that affect MWHs' health, and their experiences of health risks in LMICs using a systematic approach.
- This review will only include studies published in the English language. Excluding studies published in languages other than English may produce publication bias and impact generalizability.

BACKGROUND

The healthcare sector in low-and middle-income countries (LMICs) is growing fast [1] due to rapid population growth, and concomitant increased use of medical services [2] generates increased medical wastes[3]. Low-income countries are those with a gross national income (GNI) per capita, calculated using the World Bank Atlas method, of $1,045 or less in 2020. Lower-middle and upper-middle-income countries are those with a GNI per capita from $1,046 to $4,095 and $4,096 to $12,695, respectively [4]. According to the World Health Organisation (WHO) [5], medical wastes refer to wastes and by-products generated by healthcare and medical research settings, including medical or diagnostic centres and hospitals. While most medical wastes are not hazardous, around 15% of them is considered dangerous wastes [3 6]. These hazardous medical wastes may be infectious (contaminated blood, cultures and stocks of contagious agents), sharps (syringes, needles, disposable etc.), cytotoxic (e.g. wastes comprising cytostatic drugs—commonly utilised in cancer therapy, genotoxic chemicals), radioactive (e.g. unused liquids from radiotherapy or laboratory research, contaminated glassware, packages, or absorbent paper; urine and excreta from

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patients treated), chemical (disinfectants, sterilants and heavy metals etc.) or pharmaceutical (expired, unused and contaminated drugs and vaccines) [5-8]. These wastes contain pathogenic micro-organisms that may enter into human bodies, especially in MWHs in LMICs, through various routes, including punctures, abrasions or cuts in the skin, inhalation, and ingestion [3]. Thus, medical waste constituents have the potential for injury and infection.

Health risk can be characterised as the probability of a situation or an event and its consequences related to infectious diseases (Hepatitis B and C, HIV, and COVID-19) and other health complications (e.g. respiratory disorders, cancer, burn, and skin irritation) of individuals' (e.g. waste handlers) health [9]. MWHs, particularly those in LMICs [10], are potentially at a higher health risk to various injuries and infections due to the manual sorting of dangerous materials at waste disposal sites [3]. Compared to high-income countries, medical waste management (MWM) in LMICs is not well-equipped with the resources and capability to reduce health risks of MWHs. Additionally, the good MWM practices are not followed by waste handlers due to inadequate education and training [11]. Furthermore, the emergence of the COVID-19 pandemic has resulted in increased dangerous medical wastes from increased use of personal protective equipment (PPE) [12] such as gloves, surgical masks, goggles or face shields, gowns, respirators (i.e. KN95 or FFP2) and aprons [13]. Improper management of these used infectious medical wastes, especially in LMICs, may increase the risk of injuries and infections, including the COVID-19, Hepatitis B and C, and sharps-inflicted injuries, among MWHs [14 15].

Although studies of the individual, system and policy-levels MWM-related factors affecting the health of MWHs and their health risks in LMICs are available [16-18], the findings of these studies were not systematically synthesised. Systematically identifying, collating, synthesising, and appraising available literature will provide a comprehensive understanding of the individual, system, and policy-levels MWM-related factors affecting the health of MWHs in LMICs and their health risks and help inform future health policy and practices related to MWM in LIMCs. Furthermore, systematically synthesising various findings of available studies may help LMICs, which may not follow the best available MWM practices

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and can learn from similar settings, thus improving their MWM. This sets a strong rationale for synthesising available evidence and understanding gaps through this review and suggests recommendations for improving MWM and reducing health risks. Therefore, this systematic review will identify, appraise and synthesise qualitative evidence on the individual, system, and policy-levels MWM-related factors that affect MWHs' health in LMICs and explore common health risks experienced by them in LMICs. The review will answer the following questions: (i) what are the individual, system, and policy-levels MWM-related factors affecting the health of MWHs in LMICs? and (ii) what are common health risks experienced by MWHs in LMICs?

METHODS

This protocol is informed by the standard Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) reporting guidelines [19] (see Additional file 1). This review will utilise the ENTREQ statement to report the thematic synthesis of qualitative studies [20 21]. The ENTREQ statement aids investigators to report the phases most usually related to the synthesis of qualitative health research, including searching and selecting qualitative studies, quality appraisal, and methods for synthesising qualitative findings [20].

Patient and public involvement

No patient involved.

Search strategy

In this review, the following seven electronic databases will be searched for relevant studies in the past ten years: Scopus, Ovid MEDLINE, EMBASE, Global Health, CINAHL, ProQuest, and PsycINFO. In the case of relevant publications which might have been missed during the initial search, a further search of the bibliographical references of all eligible publications, complemented by citation tracking using Google Scholar, will be conducted. A search strategy will be developed on MEDLINE and adapted to other databases as shown in Additional file 2. The search will apply truncations (*), and Boolean operators ("AND", "OR" and "NOT") depending on the specifications of the databases. A research librarian will be consulted to finalise the search strategy. A search log will be kept for accountability and

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transparency. Two authors (MNH and TGH) will identify relevant literature, run a screening using titles, and review abstracts during the literature search. Final selection will be accomplished by reviewing full texts and applying eligibility criteria to include studies aligned with the research objectives. Database searches will be re-run prior to the final analysis to ensure any recent publications are included in the review. The search will be limited to English literature only since we have limited or no financial and logistical capacity to retrieve and translate articles published in languages other than English.

The search strategy will include all possible search terms, keywords, and phrases relevant to the topic. As per a database of interest to conduct the search, we will apply a multi-purpose (.mp) search across several fields, including title, abstract, original title, name of substance word, subject heading word, keyword heading word, protocol supplementary concept word, and unique identifier, to have efficient search outputs. The search strategy will use the following key terms: 'medical waste', 'waste management', 'training', 'knowledge', 'technology provision', 'infrastructure', 'policy', 'regulation', 'medical waste handlers', 'health risks', 'infectious disease', 'health complications', 'low and middle-income countries', 'developing countries' and 'resource-limited countries'. Along with these key concepts for search criteria, specific diseases (including Covid-19/coronavirus, HIV/AIDs, Hepatitis B and C, cancer and health complications (e.g. respiratory disorders, cancer, burn, and skin irritation) will be searched to find relevant articles.

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Inclusion and Exclusion Criteria

Types of studies

All qualitative studies published in peer-reviewed journals (1) between 1 July 2011 and 30 June 2021 with full texts available and accessible will be included in the review. Qualitative literature will be chosen because the review's interest is to understand better the experiences of MWHs' health risks. Studies (2) that apply established qualitative data collection techniques (such as semi-structured and unstructured interviews, focus group discussions, direct observations, or semi-structured questionnaires that permit free texts; (3) highlight the health risks experienced by MWHs; (4) focus on factors affecting MWHs' health risks in LMICs; and (5) published in English will be included in the analysis.

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Studies will be excluded if they (1) use statistical analysis and do not provide account/quotes of study participants, or (2) apply mixed-methods which lack accounts or quotes of study participants; and (3) are study protocols, reviews, editorials, letters to editors, commentaries, conference abstract, posters, and opinion pieces.

Participants/populations

Participants will include MWHs, defined as all workers (e.g. waste cleaners, waste pickers, collectors, recycling waste operators, scavengers, landfill workers, garbage workers) who are directly involved in medical waste collection and final disposal at waste disposal sites (city corporation bins and general landfill sites), medical research institutes, and healthcare facilities (public and private hospitals/clinics) as recommended by the WHO [8]. The participants will include: (1) MWHs of all age groups, including children from 10 years and above; (2) both male and female garbage workers; (3) waste pickers who search medical waste before or after the final disposal; (4) scavengers who search medical wastes before or after the final disposal. However, this review will exclude MWHs having severe pre-existing health risks before they became involved in MWM.

Type of setting

We will include studies from LMICs where medical waste workers are engaged in collecting MW from waste disposal sites, medical research institutes, and healthcare facilities and disposing of waste at disposal sites.

Table 1 summarises the inclusion and exclusion criteria for this systematic review.

Table 1. The inclusion and exclusion criteria for this study

Items	Inclusion criteria	Exclusion criteria
Topics	Individual, system and policy-levels factors affecting the health of MWHs in LMICs.	Individual, system and policy-levels factors affecting the health of medical waste handlers and health risks experienced by MWHs in upper-middle and high-income countries.
	Health risks experienced by MWHs.	
Types of studies	Studies that apply semi-structured and unstructured interviews, focus group discussions, direct observations, or	Protocols, reviews, editorials, letters to editors, commentaries, conference abstract and posters, and opinion pieces.

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	semi-structured questionnaires which permit free texts.	
		Studies that apply mixed-methods which lack accounts or quotes of study participants or do not provide account/quotes of study participants.
Participants	MWHs of all age groups, including children from 10 years and above; (2) both male and female garbage workers; (3) waste pickers who search medical waste before or after the final disposal; (4) scavengers who search medical wastes before or after the final disposal.	Children below 10 years of age.
		MWHs having severe pre-existing health risks before they became involved in MWM.
Settings	Waste disposal sites, medical research institutes, and healthcare facilities and disposal sites in LMICs.	Upper-middle and high-income countries.
Date range	1 July 2011 - 30 June 2021	Published prior to July 2011
Language	English	Excluded studies published in languages other than English

Outcomes

This review's primary outcome will consist of the individual, system, and policy-levels MWM-related factors affecting MWHs' health in LMICs. The secondary outcome will be the health risks experienced by the MWHs in LMICs.

DATA COLLECTION AND ANALYSIS

Study selection

All studies identified in the search will be exported into the reference manager EndNote library. A three-step screening process will be applied to screen and select eligible studies. The first step will involve assessing the titles for relevance, in which case clearly irrelevant titles will be excluded. After that, abstract screening for eligibility and relevance will be conducted, and studies that do not meet the inclusion criteria will be excluded. In the final step, the full text of the retained studies will be further screened for inclusion in the review. Final studies selected from the full-text screening will be recorded. Two reviewers (MNH and TGH) will independently perform the study selection, and any disagreements will be resolved by a third reviewer (SH).

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Data extraction and management

Data will be independently extracted by two review authors (MNH and TGH) from eligible studies onto the Joanna Briggs Institute standardised data extraction form for qualitative research [22] (see Additional file 3) and populated with variables pertaining to the study population and phenomena of interest. If there are disagreements between the two authors, a third author (SH) will double-check and verify the differences. Study characteristics that will be extracted will include the first author's name and publication year, data collection period and country in which the study was conducted. Then, descriptive data will be captured, including the study design, study population, sample size, sampling procedures and data collection procedures. The main findings and accounts of participants that explain the individual, system, and policy-levels MWM-related factors affecting MWHs' health in LMICs and the common health risks experienced by MWHs will be systematically extracted.

Quality appraisal

Studies included in this review will be assessed for methodological quality and risk of bias [23] using the Critical Appraisal Skills Programme (CASP) quality assessment tool for qualitative studies [24]. The CASP tool consists of 10 questions: 9 addressing 'quality' and 1 addressing 'value' (contribution to existing literature). All included studies will be appraised as high, medium, or low quality, and the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach [25] will be utilised to assess the overall quality of the studies included in this review. Two reviewers (MNH and TGH) will independently score the quality appraisal assessment. We will apply both scoring methods and discussion to arrive at a consensus on quality. A third author (BA) will adjudicate should no agreement be reached between the two reviewers.

In line with other qualitative reviews [26 27], this review will not exclude studies on the basis of quality. Instead, a further understanding of the contributions of the included studies will be provided at a later stage of this review. Regardless of the quality appraisal score, all studies relevant to the review questions will undergo data extraction and synthesis to assess and compare findings. The results of the quality appraisal will be reported in a narrative form and a table.

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DATA SYNTHESIS AND ANALYSIS

In accordance with existing systematic reviews [28 29], we will adopt the thematic framework analysis approach to identify the individual, system, and policy-levels MWM-related factors affecting MWHs' health in LMICs and their common health risks from the narratives of critical findings in selected studies [30]. The thematic framework analysis is one of the approaches recommended by the Cochrane Qualitative Review Methods Group [31] to perform syntheses of qualitative studies [32]. Thematic synthesis is appropriate where the evidence is likely to be mainly descriptive [28] and will enrich our understanding of the individual, system, and policy-levels MWM-related factors that can affect MWHs' health in LMICs and their experiences of health risks. We will follow five stages of the thematic framework analysis to synthesise the data [28 33] (see below).

Familiarisation with the data: The first author (MNH) will begin with familiarising the data against the review aims and documenting recurrent themes across the studies.

Identifying a thematic framework: Given our interest in identifying priori themes (individual, system, and policy-levels MWM-related factors affecting MWHs' health and their experiences of common health risks), we will use a pre-determined thematic framework which was developed using literature (see Additional file 4) to guide the thematic analysis instead of developing our own priori framework. We will also explore emergent issues (including author, country, research focus etc.) related to review objectives. This thematic framework provides a detailed list of individual, system, and policy-levels MWM-related factors affecting MWHs' health and their experiences of health risk in LMICs.

Coding: The two authors (MNH and TGH) will independently read the extracted information to search for themes according to a pre-determined thematic framework and additional emergent themes. The framework will be revised as new themes emerge. All studies will be read until there are no new emerging themes. the data will be coded based on the themes identified in the data. Each primary study will be indexed using the codes related to the themes of the framework. Where appropriate, parts of the studies may be indexed with one

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or more codes. All agreements and disparities between the coders will be resolved through discussion and consensus among all authors.

Charting: We will group the data into descriptive themes that capture and describe patterns in the data across studies and present the themes in the form of an analysis table (chart). The columns and rows of the table will reflect the studies and related themes and enable us to compare findings of the studies across different themes and subthemes.

Mapping and interpretation: We will use charts to define the identified concepts and map the range and nature of the phenomena. This systematic review will explore relationships between the themes to help clarify the findings. We will synthesise findings across studies and interpret their meanings concerning the review research questions and emerging themes.

Investigation of heterogeneity

We will assess heterogeneity of findings by conducting subgroup analyses according to: MWHs at various settings (health care facilities, incineration and waste disposal facilities, public and private healthcare facilities), ages (children versus adults), types of hazardous medical wastes (infectious, sharp, cytotoxic, radioactive, chemical, and pharmaceutical) and health risks (infectious versus non-infectious). We will also consider other axes that can emerge as essential when synthesising the evidence, such as settings (low-income versus middle-income countries) and waste management in urban and rural areas.

DISCUSSION

Globally, most LMICs have inadequate and inappropriate MWM practices, and various factors contribute to exacerbating MWM-induced health risks among MWHs [11 34]. MWHs' vulnerability to health risks and the necessity for improving MWM in LMICs highlight a research and policy priority. A substantial pool of literature exists on the subject matter but remains under-synthesised. The considerable body of literature supports the necessity for a review to provide a robust summary of evidence that could be drawn upon to optimise policies and planning related to MWM and health risks experienced by MWHs in LMICs. By analysing the account of participants across their exposure settings, and demography, this

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review could provide a direction where possible future interventions or research should focus on reducing the risks and long-term effects on MWHs. The results of this systematic review could be used to understand the individual, system, and policy-levels MWM-related factors affecting MWHs' health in LMICs and their experiences of common health risks. The review findings will inform and guide healthcare authorities on developing MWM-related interventions to improve MWHs' health in LMICs and reduce their health risks.

ETHICS AND DISSEMINATION

This review is based on published studies, which does not require ethical approval, because there is no collection of primary data. Findings from this review will be published in a reputable peer-reviewed journal and presented at a relevant public health conference. The protocol has been registered with the International Prospective Register of Systematic Reviews (PROSPERO), registration number CRD42020226851.

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Contributors

MNH, SK, and AF conceptualised the idea and designed the review. MNH, JSM, SK, SH, and AF drafted the initial protocol. MNH and SK developed the search strategy. MNH interpreted, analysed and revised the manuscripts with intellectual input from TGH, SZH, BAI, RMI and AR. All authors read and approved the manuscript.

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Competing interests or Conflict of interests

None declared.

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Additional file 1: PRISMA-P Statement—Checklist of items for qualitative studies

Items	Item numbers	Page numbers of manuscript
Title		
Identification	1a	1
Update	1b	N/A
Registration	2	2, 12
Authors		
Contact	3a	1
Contributions	3b	12
Amendments	4	N/A
Support		
Sources	5a	12
Sponsor	5b	12
Role of sponsor or funder	5c	12
Introduction		
Rationale	6	3-5
Objectives	7	2, 5-8
Methods		
Eligibility criteria	8	6-8
Information sources	9	1, 2, 5, 6, 12,
Search strategy	10	5-6
Study records		
Data management	11a	8-9
Selection process	11b	2, 8-10
Data collection process	11c	8-10
Data items	12	5-8, 12
Outcomes and prioritization	13	8
Risk of bias in individual studies	14	9
Data synthesis	15a-15d	9-11
Meta-biases	16	3, 6, 8
Confidence in cumulative evidence	17	9

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Search strategy

Additional file 2: Search terms used in this systematic review.

#	Search terms
1	(“Medical waste” OR "infectious waste" OR "hazardous waste" OR "communicable waste" OR "contagious waste" OR "dangerous waste" OR “risky waste”, OR “unsafe waste” OR "transmittable waste” OR “epidemic waste” OR “spreading waste” OR “biomedical waste” OR “clinical waste” OR “biohazardous waste” OR “healthcare waste” OR “sharp waste*” OR “radioactive waste” OR “pathological waste *” OR “pharmaceutical waste” OR “laboratory waste” OR “chemical waste”).mp. [mp=Title, Abstract, Original Title, Name of Substance Word, Subject Heading Word, Keyword Heading Word, Protocol Supplementary Concept Word, Unique Identifier]
2	(“Manage* OR “disposal* OR “regulat*” OR “separate*”, OR “stor*”, OR “transport*” " intervention*” OR “process*" OR " handling*" OR " discharg*" OR “recycle*”).mp. [mp=Title, Abstract, Original Title, Name of Substance Word, Subject Heading Word, Keyword Heading Word, Protocol Supplementary Concept Word, Unique Identifier]
3	(Factor* OR determinant* OR issue* OR reason* OR cause* OR training OR knowledge OR infrastructure OR “technology provision” OR policy OR regulation).mp. [mp=Title, Abstract, Original Title, Name of Substance Word, Subject Heading Word, Keyword Heading Word, Protocol Supplementary Concept Word, Unique Identifier]
4	(Health OR “physical health” OR well-being OR wellness).mp. [mp=Title, Abstract, Original Title, Name of Substance Word, Subject Heading Word, Keyword Heading Word, Protocol Supplementary Concept Word, Unique Identifier]
5	(“Waste handler” "waste picker*" OR "garbage worker*" OR "rag picker*" OR scavenger*" OR " waste cleaner*" OR "landfill worker*" OR "informal worker*" OR "recycler*" OR "sanitation worker*" OR "trash collector*" OR “waste operat*” OR "sewerage worker*").mp. [mp=Title, Abstract, Original Title, Name of Substance Word, Subject Heading Word, Keyword Heading Word, Protocol Supplementary Concept Word, Unique Identifier]
6	(“Health risk*” OR “health effect*” OR “health complication*” OR “communicable disease*” OR “infectious disease*” OR “contagious disease*” OR “contagious illness*” OR “transmissible disease” OR “physical risk” OR “respiratory disorder*” OR “liver disease” OR COVID-19 OR coronavirus OR cancer OR HIV OR AIDS OR hepatitis OR typhoid OR

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	burn OR irritation, OR injury).mp. [mp=Title, Abstract, Original Title, Name of Substance Word, Subject Heading Word, Keyword Heading Word, Protocol Supplementary Concept Word, Unique Identifier]
7	("Low-income countr*" OR "middle-income countr*" OR "low and middle-income countr*" OR "resource-constrained countr*" "resource-limited countr*" OR "developing countr*" OR "marginalised countr*" OR "underprivileged countr*" "poor countr" OR "unindustrialised countr*" OR "emerging countr**" OR "less developed countr*" OR "underdeveloped countr*").mp. [mp=Title, Abstract, Original Title, Name of Substance Word, Subject Heading Word, Keyword Heading Word, Protocol Supplementary Concept Word, Unique Identifier]
8	#1 AND #2 AND #3 AND #4 AND #5 AND #7 Filters: Full text; English, NOT systematic review, NOT quantitative studies.
9	#1 AND #5 AND #6 AND #7 Filters: Full text; English, NOT systematic review, NOT quantitative studies.

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Additional file 3: JBI data extraction tool

JBI QARI Data Extraction Tool for Qualitative Research

Reviewer _____ Date _____

Author _____ Year _____

Journal _____ Record Number _____

Study Description

Methodology

Method

Phenomena of interest

Setting

Geographical

Cultural

Participants

Data analysis

Authors conclusions

Comments

Complete Yes ☐ No ☐

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Findings	Illustration form Publication (page number)	Evidence		
		Unequivocal	Credible	Unsupported

Extraction of findings complete Yes ☐ No ☐

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Additional file 4: Thematic framework analysis format for summarizing factors influencing MWM and health risks experienced by medical waste handlers in LMICs.

Main themes	Emerging sub-themes	Studies
Factors influencing MWM in LMICs		
Health risks experience		

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